# Mapiah

User guide

1.0.2

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# Installation

# Installation instructions

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## Introduction

This document provides detailed instructions for installing Mapiah on the three supported operating systems: Linux, macOS, and Windows.

Since Mapiah is software intended to make working with Therion more user-friendly, the first step is to install Therion on your system. The second step is to install Mapiah itself.

## Linux

### Installing Therion on Linux

Therion packages for several Linux distributions: Debian GNU/Linux, Arch Linux, Ubuntu, Fedora.

After installing Therion, test the installation as described in the "First run of Therion (test)" section below.

### Installing Mapiah on Linux

For Linux there are two available Mapiah builds: an AppImage file and a Flatpak file. Both provide the same functionality, but the AppImage is simpler to use, while Flatpak downloads are usually smaller, especially on future updates.

#### Option 1: Install Mapiah via AppImage

1. Find the latest available Mapiah version on the Mapiah releases page.
2. Download the corresponding `.AppImage` file.
3. Make the downloaded file executable, for example:

```
chmod +x Mapiah-<version>.AppImage
```

4. Run Mapiah:

```
./Mapiah-<version>-linux-x86_64.AppImage
```

5. In Mapiah, open the `cave.th2` file from Therion's sample data (see the "First run of Therion (test)" section below).

   *Note:* If your Linux installation does not support `.AppImage` files, look up how to enable AppImage support for your specific distribution.

#### Option 2: Install Mapiah via Flatpak

1. If you have the old Flathub package installed, remove it:

```
flatpak remove io.github.rsevero.mapiah
```

2. Install Flatpak on your system if it is not already available. Instructions are available at: https://flatpak.org/setup/
3. Install Mapiah from the Mapiah Flatpak repository:

```
flatpak install --user --from
https://flatpak.mapiah.org/org.mapiah.mapiah.flatpakref
```

4. Update it with:

```
flatpak update org.mapiah.mapiah
```

# macOS

## Installing Therion on macOS

To install Therion on macOS, follow the instructions below (adapted from the homebrew-therion repository page).

Please open the Terminal app and follow these instructions:

1. Install the command line tools:

   xcode-select --install

2. Install Homebrew - http://brew.sh/

   /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

Test the installation with:

   brew update brew doctor

3. Install XQuartz If your macOS does not include the X11 server (10.12 Sierra+), install it with:

   brew install --cask xquartz

4. Install MacTeX

   brew install --cask mactex

5. Install tcl-tk and bwidget

   brew install tcl-tk bwidget

6. Install Therion

   brew tap ladislavb/homebrew-therion brew install therion

7. Copy Loch to /Applications Loch is installed at `<prefix>/opt/therion/loch.app/`. If you have an older Loch version installed in Applications, remove it and copy the new version with:

   cp -R /opt/therion/loch.app /Applications/loch.app

Replace `<prefix>` with `/usr/local` for Intel macOS or `/opt/homebrew` for Apple Silicon.

After a successful installation, you should be able to:

```
start XTherion by typing the `xtherion` command in Terminal
run the Therion compiler by typing the `therion` command in Terminal
start the Loch viewer from Launchpad
```

### Installing Mapiah on macOS

Mapiah requires macOS 12 (Monterey) or later.

1. Find the latest available Mapiah version on the Mapiah releases page.
2. Download the corresponding `.dmg` file.
3. Open the downloaded `.dmg` file and drag the Mapiah icon to the Applications folder.
4. Run Mapiah from the Applications folder.
5. In Mapiah, open the `cave.th2` file from Therion's sample data (see the "First run of Therion (test)" section below).

## Windows

### Installing Therion on Windows

Download the Therion installer for Windows from the Therion downloads page and follow the installer's instructions.

### Installing Mapiah on Windows

1. Find the latest available Mapiah version on the Mapiah releases page.
2. Download the corresponding `.exe` file.
3. Run the downloaded `.exe` and follow the installer's instructions.
4. After installation, start Mapiah from the Start menu.
5. In Mapiah, open the `cave.th2` file from Therion's sample data (see the "First run of Therion (test)" section below).

## First run of Therion (test)

After installing Therion, you can test it with the sample data available on the Therion website:

1. Download the sample data from the Therion website and unzip it somewhere on your disk.
2. Run XTherion (on Unix and macOS, type `xtherion` on the command line; on Windows, there is a shortcut in the Start menu).
3. Open the `thconfig` file from the sample data directory.
4. Press F9 or click `compile` in the menu to run Therion on the data — you will see some messages from Therion, MetaPost, and TeX.
5. PDF maps and a 3D model will be created in the sample data directory.

## Updating Mapiah

To update Mapiah, follow the same installation instructions for your operating system, downloading the latest available version from the Mapiah releases page.

# Workspace and projects

Initial workspace page where the project tree, file tabs, and main actions are presented.

The complete printable documentation, including installation instructions, help pages, and keyboard shortcuts, is in `Mapiah-User-Guide-en.pdf` at the root of the Mapiah repository. A copy is attached to every release.

*Note: Mapiah treats the Ctrl and Meta (Command on macOS) keys as interchangeable. Shortcut mentions below use "Ctrl" for brevity.*

## Project workflow

**Open project** is the normal entry point for a Therion project. Select any root configuration filename, including a configuration without the `.thconfig` extension. Mapiah loads that file and recursively follows its `source`, `input`, and related references into the project tree. Opening another project replaces the loaded tree; the usual file-tab lifecycle is preserved.

The sidebar contains file nodes and logical nodes. File nodes represent `thconfig`, `.th`, and `.th2` source files. Logical nodes represent scraps, surveys, maps, and centrelines discovered in those files. Use the search field to filter nodes and the chevrons to expand or collapse branches. The sidebar width and collapsed state are persisted in settings. A dirty marker means a text file has unsaved changes. Error indicators identify parser or compiler diagnostics; missing files and circular references are reported as project errors.

Click a `.th2` file node to open its canvas tab. Click a `thconfig` or `.th` node to open its text tab. Clicking a scrap, survey, map, or centreline selects the corresponding source and navigates to its location when a source line is available.

Expanding a `.th2` file row loads its scrap and element list without opening a tab; only the files you expand are loaded. The list shows the file's drawing order. See the **Drawing order and element tree** and **Broken files** sections of the TH2 file edit help for details.

Use **Run Therion** to run the loaded project's root configuration. In the empty state, the project-tree Run Therion action opens a project and runs it. With a project loaded, the action reruns that project. Compiler diagnostics replace the previous run's compiler diagnostics after the next run; editing alone does not clear them.

## Standalone `.th2` files

Standalone drawings can be opened by clicking a `.th2` file node in a loaded project, or at startup with a positional path or the `--th2` option. `Ctrl/Cmd+O` opens a project in the workspace; it is not a standalone-file picker.

```
mapiah file.th2
mapiah --th2 file.th2
```

## Top bar

- **Open project**: opens a project configuration in the workspace. `Ctrl/Cmd+O` and `Ctrl/Cmd+Shift+O` use this project-opening action.

- **Run Therion**: runs the loaded project, or opens a project and runs it when no project is loaded.
- **Settings page**: opens application settings.
- **Keyboard shortcuts page**: shows the available shortcuts.
- **Help**: shows this dialog.
- **About**: shows application information.

## Command-line Arguments

Mapiah supports command-line arguments to open files directly when starting.

### Positional Arguments

```
mapiah /path/to/file.th2          # Opens a standalone TH2 canvas
mapiah /path/to/therion.cfg       # Loads the project and runs Therion
```

Mapiah detects TH2 files by their `.th2` extension and treats any other file as a project configuration. The selected configuration becomes the root of the project tree.

### Named Arguments

#### --th2: Open standalone TH2 files

This option can appear multiple times; each file opens in a separate canvas tab.

```
mapiah --th2 file1.th2 --th2 file2.th2
mapiah --th2 /path/to/survey.th2
```

#### --thconfig: Load and run a project

Use at most one `--thconfig` per command. The project tree is loaded for the selected configuration, whose filename does not need to end in `.thconfig`.

```
mapiah --thconfig project.cfg
mapiah --thconfig /path/to/therion.cfg
```

#### --therion_run_parameters: Set Therion command-line options

Sets extra options passed to Therion when compiling (for example, `-d` for debug mode). The value is persisted as the `Main_TherionRunParameters` setting.

```
mapiah --therion_run_parameters "-d -q"
mapiah --thconfig project.cfg --therion_run_parameters "-d"
```

If a required flag value is missing, or more than one `--thconfig` is provided, Mapiah exits with an error.

# Drawing editor

This is where all TH2 file editing is done.

*Note: Mapiah treats the Ctrl and Meta (Command on macOS) keys as interchangeable. Shortcut mentions below use "Ctrl" for brevity.*

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## Top bar

- On the left:
    -  *Back*: returns to the main window without saving contents.
- On the right:
    -  *Save*: saves changes in same file. Only enabled if there are changes to be saved. (Ctrl+S)
    -  *Save As*: saves changes in new file. (Shift+Ctrl+S)
    - *Run Therion*: when no project is loaded, opens a project and runs it; otherwise, reruns the loaded project.
    -  " *Run Therion*: runs Therion with currently opened project.
    -  *Help*: show this dialog box.
    -  *Close*: close the TH2 file edit window without saving changes.

## File tabs

When multiple files are open, each file appears as a tab at the top of the editor. The file name is displayed on the tab along with an **X** button to close that file.

**Tab features:**

- **Switch between files**: Click on any tab to switch to that file
- **Close a file**: Click the **X** button on the tab to close that file
- **Scroll tabs**: If you have many files open, use your mouse to click and drag horizontally on the tabs to scroll through them
- **Multiple selection**: Use the *File open* button on the top bar to select and open multiple files at once

The currently active file's tab is highlighted, making it easy to see which file you're editing. All open files maintain their editing state, so you can switch between them without losing your work.

## Project text editor

Project `thconfig` and `.th` files open in text tabs. Syntax highlighting and folding make the source easier to read. While you edit, parsing is debounced and the project tree is refreshed after the pause. The tab becomes dirty until you save or revert it. *Ctrl+Z* undoes and *Ctrl+Shift+Z* redoes text edits, as on the canvas. Find and replace apply to the active file only; project-wide search is not available.

Parser diagnostics are produced while loading or re-parsing source and are shown in the editor and project tree. Compiler diagnostics are produced by running Therion. Both kinds can coexist. A diagnostic with a known file and line can be selected to navigate to that location; a diagnostic without a source line remains in the run output and cannot target a tree line.

## Drawing order and element tree

Each `.th2` file row in the project tree has an arrow. Expanding it loads the file and shows its scraps and, below each scrap, its points, lines and areas. Comments, empty lines, settings, images and line points are not rows; they stay where they are in the file.

- **Loading without opening a tab**: expanding a file reads it in the background. The tree shows *Loading…* until it is ready. No tab opens and nothing is changed. Only files you expand are read; files are never read just to answer a search.
- **Drawing order**: rows are in file order, which is the order XTherion uses. The top row is drawn first (bottom of the stack) and the last row is drawn last (on top). Therion's own rendered layering may differ, because it also depends on the symbol types. On Mapiah's canvas, selected elements are temporarily painted above the others.
- **Areas**: an area's fill is painted together with its border line, so changing only the area row's order may have no visible effect on the canvas. Move the border line's row to change where the fill is layered.
- **Labels**: each row shows the element kind, its type (and subtype, if any), then, in italics, a station's name or the text of a label or remark point, and then its Therion id, if it has one, exactly as written in the file. Hover over a multi-line label or remark to see all its lines.
- **Type icons**: the icon of each point, line and area row is a small preview of its type and subtype, drawn as the canvas draws it with the current *Visualization method* setting, on the canvas background color. Lines and areas are shown in a small frame on a fixed sample shape; an area's preview shows only its own fill and outline, not the symbols of its border lines. The icons depend only on the type and subtype: zoom, the symbol size setting, the scrap scale, other options and the element's shape do not change them. A label point shows its point marker, not its text.
- **Scraps**: scraps start collapsed, so expanding a file shows only its scraps. Click the arrow of a scrap to collapse or expand it; collapsing a scrap hides its rows. A collapsed scrap shows a dot when one of its elements is selected.
- **Selecting**: clicking a point, line or area selects it, as the Select tool followed by a click on the element would. If the file already has a tab, that tab is brought to the front. If it has no tab, the element is selected without opening one. Clicking a scrap makes it the active scrap.
- **Selecting several rows**: Ctrl+click a point, line or area to toggle it in the active scrap's selection. Shift+click extends the selection from the last clicked element row. Ctrl+click scraps selects several scraps for reordering without changing the active scrap; a plain click on a scrap only makes it active. Selection cannot span scraps.
- **Double-click**: double-clicking an element also opens the file's tab if needed and zooms to the selection.
- **Selection follows the canvas**: the rows of elements selected on the canvas are highlighted, and the active scrap's row is highlighted too.
- **Dragging**: drag selected rows to move them together in file order. On a point, line or area row, the upper half places them before it and the lower half after it. On a scrap row, points, lines and areas go to the start or end of that scrap; scraps go before or after the whole scrap. The upper and lower halves of a file row mean the start and end of its scrap list. A collapsed scrap expands after a short hover. The tree scrolls when dragging near its edge. Dragging is unavailable while search is active.
- **Refused moves**: a mixed selection of scraps and drawing elements and a move between files are refused. Moving an area to another scrap carries its border lines along; a move that would leave a

border line in a different scrap from any area that uses it is refused. The drag preview shows why a drop is refused, and unavailable Move to scrap targets show why below their names.
- **Order menu**: right-click an element row for Bring forward, Send backward, Bring to front or Send to back. Points, lines and areas also have Move to scrap. A right-click selects an unselected drawing element; a selected row keeps the selection. These actions move whole elements among their siblings, including the internal content of a line or area.
- **Order shortcuts**: Ctrl+] brings the selection forward, Ctrl+[ sends it backward, Ctrl+Shift+] brings it to the front and Ctrl+Shift+[ sends it to the back (Cmd instead of Ctrl on macOS). They act in the active canvas tab on the selected drawing elements, or on the scraps selected in the tree, and need keyboard focus in that tab. They are ignored while Alt is pressed.
- **Undo**: every tree edit opens the file's tab if needed and can be undone with Ctrl+Z. Tree edits are disabled in text mode (see Text mode).
- **Search**: the project tree search field also matches the full labels of the scraps and elements of files that are already loaded and valid, including station names and label or remark text. Matching rows are shown together with their scrap and file.
- **Closing a tab**: closing the tab of a file whose tree row is expanded keeps its elements in the tree, unless the file had unsaved changes, which are discarded as usual.

## Broken files

When a `.th2` file has any parsing or structural problem, Mapiah marks it as broken. Its element rows and canvas editor are unavailable. Fix it with *Edit as text* (or F2) on its page, or in another text editor and reload it.

Examples of problems:

- A point, line or area outside a scrap.
- A scrap inside another scrap.
- An `endscrap` without an open scrap, or a missing `endline`, `endarea` or `endscrap`.
- An area border that does not refer to a valid line.
- A malformed command, an unknown command or an unknown option.

Unknown point, line or area types alone are still accepted; unknown options are errors.

- **In the project tree**: the file row shows a badge with its number of problems. Hover over it to see the line number and category of the first problems. Expanding the row shows a single row explaining that the file must be fixed with *Edit as text*, or outside Mapiah and reloaded.
- **In a tab**: opening a broken file, from the project or on its own, shows a page instead of the canvas. It explains the situation, shows the file path with a *Copy path* button, and lists every problem with its line number, its category and the source line. *Details* reveals the original parser message, which is useful when reporting a problem.
- **Edit as text**: shows the file's text as it is on disk, with the problems marked on their lines and the cursor on the first one. A file that can't be read as text, or isn't valid in its encoding, can't be opened this way; the message says what to fix. See Text mode for how text editing works.
- **Applying the fix**: once the text has no problems, applying it (by leaving text mode or saving) makes the canvas available. This first apply is not an undo step: undo can't bring back the broken file. The fix stays unsaved until you save; closing or reloading the tab discards it without asking, as for canvas edits.
- **Leaving text mode**: with text that fixes the file, leaving shows the canvas; with edited text that still has problems, it shows the "can't apply" banner; with the broken text untouched, it returns to the broken-file page. While the file is broken, the text view's button says *Leave text mode*.
- **Reload**: after fixing the file outside Mapiah, use the *Reload* button on that page, or right-click the file row or its explanation row in the tree and choose *Reload*. Reload brings back the file as it is on disk. The

same tree option applies to a file whose loading failed unexpectedly.
- **Running Therion**: when you run the loaded project, the *Run Therion* dialog warns about the project's loaded files that are broken on disk, because Therion may fail on them. The run still starts. A fix applied in text mode but not saved yet still counts as broken, because Therion reads the file on disk; the warning marks it "(fix not saved)". The warning only covers loaded files: after saving a fix with *Save As* under a new name, the original file, still broken on disk, can break the run without a warning.

## Text mode

A `.th2` tab can show its file as Therion source and be edited as text.

- **Switching**: the *Text* button in the top right corner, or F2, shows the file as text; the text view's *Canvas* button, or F2, goes back to the canvas. The canvas buttons, the last-used chips and the status bar are hidden in text mode. Switching to a tab that is in text mode puts the focus in the text. If an image is being inserted, the switch waits for it to finish. A file with a carriage return that isn't followed by a line feed can't be opened in text mode: normalize its line endings in another editor and reload it.
- **Selection**: entering text mode puts the cursor on the first selected element, in file order. Leaving it selects the element on the cursor's line: a point, line or area is selected; a line point opens line edit with that point selected; a scrap line makes that scrap active; a file-level line clears the selection. The canvas is centered on the selection without zooming.
- **Applying**: leaving text mode or saving with changed text checks the text and, if it has no problems, applies it to the drawing as a single undo step. If it has problems, nothing is applied, the problem lines are marked and a banner offers *Keep editing* and *Discard text changes*. Characters that the file's encoding can't hold, text Mapiah can't read and duplicate IDs are problems too. *Discard text changes*, on the banner or the text toolbar, drops the text edits and returns to the canvas with the file unchanged.
- **Markers**: while you edit, information markers show the lines that saving will rewrite or remove (duplicate line points, lines with fewer than two points, areas without a border), the settings that will move up into the settings section (the marker says to which line), and the lines or line endings that will be added. They don't prevent applying the text.
- **Undo**: before your first text edit, Ctrl+Z and Ctrl+Shift+Z undo and redo canvas edits, and the text follows. From the first text edit on, they act on the text only, and canvas edits can't be reached until you leave text mode. Typing never deletes canvas history. Each apply is one canvas undo step.
- **Saving**: *Save* checks and applies the text first; text with problems is never written. After saving, the text shows exactly what was written and its undo history starts again. *Save As* checks the text before the file dialog opens and applies it after a name is chosen; cancelling changes nothing, and the text can't be edited while the dialog is open. If the text editor can't be reopened after a save, or after an undo or redo of canvas edits, Mapiah returns to the canvas with a message; nothing is lost and F2 opens the text again.
- **Settings section**: comments and empty lines between the settings at the top of the file are kept. Settings found later in the file move up into the settings section, before its trailing empty lines. Ordinary saving from the canvas follows the same rule (see Original file format).
- **The project tree**: a click or right-click on a row moves the text cursor to that element or scrap, following it through your edits, and doesn't change the canvas selection or the active scrap. Drag and drop, drawing order, *Move to scrap*, *Cut*, *Duplicate*, *Delete*, *Add scrap*, *Options…* and the tab's *Properties* are disabled; go back to the canvas to use them. The encoding is changed on the `encoding` line. *Copy* is also offered on point, line and area rows, and copies TH2 source that can be pasted into the text. IDs are copied as they are, so pasting into the same file gives duplicate IDs that must be fixed. Scrap visibility works as on the canvas. While Mapiah is applying or saving the text, tree clicks are ignored.
- **Find**: Ctrl+F finds and replaces in the text, as in `thconfig` and `.th` tabs. Multi-file search doesn't include `.th2` files.

# Edit window

## Top right corner

- *Text*: shows the file as text (see Text mode). (F2)
- *Search*: opens the search and select dialog. (See Search and select)
-  *Snap*: toggles the snap window where are presented the snap options. (Ctrl+L)
- *Default options*: opens the default options overlay window where default option values can be set for new points, lines, and areas. (O with no elements selected)
-  *Delete*: deletes the currently selected elements. Only enabled if there is at least one element selected. (Delete/Backspace)
-  *Undo*: undos the last executed edit operation. Only enabled if there is at elast one edit operation to be undone. (Ctrl+Z)
-  *Redo*: redoes the last undone edit operation. Only enabled if there is at least one edit operation to be redone. (Ctrl+Shift+Z)

In case there are available redoes when a new edit operation is performed, the redo stack is migrated to the undo stack making redoes still accessible.

## Bottom right corner

-  *Images*: opens the images options overlay window. Shows all images inserted in the current file. Each image row has a visibility checkbox, a grid visibility checkbox (XVI images only), a delete button, and a drag handle for reordering. Also presents an "Add Image (I)" button. (Alt+I)
-  *Select element*: allows to select elements in the TH2 file. (C)
-  *Line edit*: allows to edit individual lines in the TH2 file. (N)
  - Double-clicking a line or one of its visible line segments while using the *Select element* tool also enters *Line edit* mode for that line.
-  *Add element*: allows to add new elements to the TH2 file. On mouse over the following buttons are shown:
  -  *Add point*: adds a new point to the TH2 file. (P)
  -  *Add line*: adds a new line to the TH2 file. (L)
  -  *Add freehand line*: draws a new line by dragging the pointer; the stroke is simplified and stored as a series of straight segments. (F)
  -  *Add area*: adds a new area to the TH2 file. (A)
-  *Zoom options*: presents the zoom options. On mouse over the following buttons are shown:
  -  *Zoom in*: zooms in the TH2 file view. (+)
  -  *Zoom one to one*: zooms the TH2 file view to show the elements at their original size, i.e., each TH2 point corresponding to a screen pixel. (1)
  -  *Zoom to selection*: zooms the TH2 file view to show the currently selected elements. (2)
  -  *Zoom to selection window*: zooms the TH2 file view to show the window selected by the user with a mouse drag. (5)

-  *Zoom to file*: zooms the TH2 file view to show all elements in the file. (3)
-  *Zoom to scrap*: zooms the TH2 file view to show the currently selected scrap. (4)
-  *Zoom out*: zooms out the TH2 file view. (-)

## Drawing lines

When drawing lines, each new segment is initially created as a straight line segment. To convert it to a Bézier Curve line segment, do not release the mouse button and drag. The mouse position will be treated as the position of the single control point of a quadratic Bézier Curve.

The exact drag behavior depends on the settings option "New line creation method". "Mapiah quadratic" keeps the current behavior described below and also lets you hold *Shift* during the drag to constrain the control point to the nearest multiple of the configured snap angle relative to the shared node. "xTherion cubic smooth" uses the dragged position as the next segment's future control point, mirrors the current segment's other control point around the shared end point, and lets you hold *Ctrl* while dragging to keep the mirrored control point at a fixed distance. While setting that control point in "xTherion cubic smooth" mode, you can also hold *Shift* during the drag to constrain it to the nearest multiple of the configured snap angle relative to the shared node. This *Shift* constraint can also be combined with *Alt* in "xTherion cubic smooth" mode.

When drawing with either "Mapiah quadratic" or "xTherion cubic smooth", you can also hold *Shift* while setting a node to constrain it to the nearest multiple of the configured snap angle relative to the previous node.

When drawing with "xTherion cubic smooth", you can also make the shared node a corner point by separating the two control points. Start dragging normally to define the current segment's control point at the shared end node. Then, without releasing the mouse button, hold *Alt* and keep dragging to redefine only the next segment's future control point. This keeps the current segment handle fixed while changing the next one independently.

When drawing with either "Mapiah quadratic" or "xTherion cubic smooth", you can press *Backspace* or *Delete* to remove the last created node. If that was the only completed segment so far, Mapiah returns to the original start node so you can continue drawing from it.

When drawing with either "Mapiah quadratic" or "xTherion cubic smooth", you can press *Esc* to delete the whole unfinished path and start a new one.

When drawing with either "Mapiah quadratic" or "xTherion cubic smooth", you can press *Shift+L* to convert the last (unfinished) segment from a Bézier curve to a straight line, or *Shift+U* to convert it from a straight line to a Bézier curve. When converting to a Bézier curve, the control points are placed at one-third and two-thirds of the segment chord.

While drawing a line, you can also move the last created node with the keyboard:

- Press an *Arrow* key to move it by the configured nudge factor (`TH2Edit_NudgeFactor`), measured in canvas pixels
- Press *Shift+Arrow* to move it by ten times the nudge factor
- Press *Alt+Arrow* to move it by 1 screen pixel
- Press *Alt+Shift+Arrow* to move it by 10 screen pixels

Bézier Curves on Therion (and Mapiah) are cubic curves, i.e., they have 2 control points for each segment. Just on line segment creation Mapiah pretends that the Bézier Curve being created is a quadratic Bézier Curve (with only one control point) so the user has flexibility to create the line segment.

Observe that despite the fact that Mapiah is simulating the existance of only one control point, an actual cubic Bézier Curve is being created with two control points as expected.

### Map connection

For selected X-Section points, on Ctrl+X, Mapiah searchs for a scrap option that ends on "-xs-STATION" where STATION is the station where the x-section was created. It searchs for a point type station with its name option set to STATION. If both are found, a line of type "map connection" is created between the x-section point and the station point.

## Selecting elements

To select an element, click on it with the *Select element* tool active. To select multiple elements, hold the *Shift* key while clicking on them. To deselect an element, hold the *Shift* key while clicking on it. To select all elements, use the *Select all* option on the *Select element* tool or use the keyboard shortcut *Ctrl+A*.

Its also possible to select elements by dragging a selection window with the mouse. To do that, click and hold the left mouse button on an empty area of the canvas and drag the mouse. All elements that are fully or partially inside the selection window will be selected. To add elements to the selection, hold the *Shift* key while dragging the selection window.

When clicking on a line that defines an area border, the behavior depends on which modifier keys are held:

1. **No modifier (or modifiers other than Ctrl+Shift)**: both the area and the line are candidates for selection. If exactly one is already selected the other is added; otherwise a "Multiple elements clicked" dialog is shown so you can choose which element(s) to add to the selection.
2. **Ctrl+click (without Alt or Shift)**: only border lines are added to the selection directly, without showing the dialog. If the area has more than one border line, the first click selects all of its border lines. Additional Ctrl clicks, while you keep Ctrl held, cycle through the same area's border lines in the order they appear in the area's `THAreaBorderTHIDs`, then return to selecting all border lines again.
3. **Ctrl+Alt+click (without Shift)**: only the area is added to the selection directly, without showing the dialog.

## Element operations

### Copy and paste

Selected elements can be copied to a clipboard and pasted into the current file or another open file.

**To copy selected elements:**

- Press *Ctrl+C*
- At least one element must be selected
- **To paste copied elements:**
- Press *Ctrl+V*
- Pasted elements appear in the current scrap
- All child elements (line segments, area borders, etc.) are automatically included in the paste

- THID references are automatically resolved to avoid conflicts: if a pasted element's THID already exists in the target file, a new unique THID is automatically generated
- Pasted elements become the new selection, making them ready for further editing or moving
- The paste operation can be undone with *Ctrl+Z*

**Cross-file pasting:** When you have multiple files open in tabs, you can copy elements from one file and paste them into another file. Simply switch to the target file tab and press *Ctrl+V*.

### Cut

Selected elements can be cut (copied to clipboard and immediately removed from the file).

- **To cut selected elements:**
- Press *Ctrl+X*
- At least one element must be selected
- The elements are copied to the clipboard and then removed from the file
- The clipboard content can be pasted with *Ctrl+V* into the same or another open file
- The cut operation can be undone with *Ctrl+Z*, which restores the elements to their original positions

### Duplicate

Selected elements can be quickly duplicated in place.

**To duplicate selected elements:**

- Press *Ctrl+D*
- All selected elements and their children are duplicated
- Duplicated elements appear at the same position as the originals
- The duplicate operation creates new unique IDs for all duplicated elements
- Duplicated elements become the new selection
- The duplicate operation can be undone with *Ctrl+Z*

## Element options

Right clicking on a selected element presents an overlay window with the options available for the currently selected elements. The element options window can also be opened by using the 'O' keyboard shortcut when there is at least one element selected. The options available depend on the type of element selected.

To edit scrap options, right-click its row in the project sidebar and choose **Options…**.

The scrap *Sketch* option sets the background images Therion draws under the scrap when the layout has `sketches on`. After choosing *Set*, every sketch of the scrap is listed with its filename and lower left corner coordinates, which can be edited directly; use *Remove sketch* to delete one. *Add loaded image* adds a sketch from a raster image already loaded in the file, filling in its filename and lower left corner coordinates. *Add from file…* adds a sketch from an image file on disk; as that image isn't placed on the canvas, type in its lower left corner coordinates. Chosen files are stored relative to the TH2 file. Therion doesn't scale or rotate sketches, so a warning is shown when the loaded image is scaled or rotated in Mapiah.

# Default options

Default options are option values that are automatically applied to newly created points, lines, or areas of matching types. They are set once and reused across all subsequent element creations until changed or removed.

Open the default options overlay window by:

- Pressing 'O' with no elements selected, or
- Clicking the *Default options* button (tune icon) in the top right corner.

The overlay window contains three tabs: **Points**, **Lines**, and **Areas**. Select the tab for the element category whose defaults you want to configure.

Within each tab, the available options work identically to the regular element options editor. Setting an option stores it as a default for the selected category. Unsetting an option removes it from the defaults.

Only options applicable to the specific type of the newly created element are applied. For example, if a default "clip" option is set for lines, it will only be applied when creating line types that support the "clip" option.

A **Reset** button at the top of the overlay clears all defaults for the currently shown tab. The button is disabled when no defaults are set for that category.

# Save

## Original file format

Mapiah preserves the original file formatting as much as possible when saving. However, some changes are made on file parsing that are reflected in the saved version even if the user did no editing at all:

- Consecutive line points with identical end points are merged into a single line point.
- Lines with zero or one line points are removed.
- Non existing area border references are removed, i.e., the area mentions an area border ID but there is no line with the same ID on the file.
- Areas with no border references are removed.
- Line options defined in [LINE DATA] area are moved to the line definition. Not to be confunded with line point options that are defined in [LINE DATA] area and are preserved there.
- Subtype options defined before the first line point or on the first one are transformed to line subtype.
- The settings section runs from the `encoding` line to the first element that isn't a setting, a comment or an empty line. Comments and empty lines between settings are kept. Settings found later in the file are moved up into the settings section, before its trailing empty lines; nothing else moves.

# Images

The images overlay window is opened with the  button (Alt+I) in the bottom right corner. It lists all images (XVI survey backgrounds, raster images, and Mapiah-only SVG images) inserted in the current file.

SVG images are always stored as Mapiah-only image inserts. To import an SVG, the file must define either a `viewBox` or numeric `width` and `height`. If neither is available, Mapiah shows an import error and does not insert the image.

A *toggle all* button appears above the list when images are present. Its tooltip and icon reflect what the button will do:

- *Hide all images* (eye-off icon): shown when all images are visible; clicking hides all images.
- *Show all images* (eye icon): shown when any image is hidden; clicking makes all images visible.

Each image row contains:

- A visibility checkbox to show or hide the image on the canvas
- A grid visibility checkbox (XVI images only) to show or hide the survey grid independently from the image itself
- The image filename
- An edit button to enter image transform mode on the canvas
- A reset button to restore image translation, scale, and rotation to their defaults
- A delete button to remove the image
- A drag handle (⣿) to reorder images

## Image visibility

Clicking the visibility checkbox toggles whether the image is displayed on the canvas. Hidden images are still stored in the file.

## Image transform mode

Click the edit button on an image row to enter image transform mode for that image. The image is outlined on the canvas with small black handles on the corners and on the middle of each side. Clicking the selected image again while transform mode is active switches to image rotation mode, where the corner handles become curved rotation handles and a pivot marker is shown.

While image transform mode is active:

- Drag the image itself to move it
- Drag any black handle to scale it
- Click the selected image to toggle between move/scale mode and rotate mode
- Press *Alt+M* to mirror the image horizontally
- Press *Ctrl+M* to mirror the image vertically
- Use the two mirror buttons shown on the left side of the canvas to trigger the same actions with the mouse
- Press *Esc* to leave image transform mode

Click the reset button on an image row to set `xx`, `yy`, and the rotation angle back to `0`, `xScale` and `yScale` back to 1, and image visibility and XVI grid visibility back to their default visible state. The reset keeps the XVI root unchanged.

## Image moving

In image transform mode, drag the image body to move the selected image.

Keyboard movement:

- Press an *Arrow* key to move the selected image by the configured nudge factor (`TH2Edit_NudgeFactor`), measured in canvas pixels
- Press *Shift+Arrow* to move by ten times the nudge factor
- Press *Alt+Arrow* to move by 1 screen pixel

- Press *Alt+Shift+Arrow* to move by 10 screen pixels

The following modifiers can be combined while moving:

- Hold *Alt* and drag anywhere on the canvas to move the selected image without needing to start on the image body
- Hold *Ctrl* while dragging to constrain the move to the dominant horizontal or vertical direction
- Hold *Shift* while dragging to temporarily disable snapping

## Image scaling

In image transform mode, drag any black handle to scale the selected image.

You can also mirror the image instantly without dragging:

- Press *Alt+M* to negate `xScale`
- Press *Ctrl+M* to negate `yScale`
- The mirror actions are undoable with *Ctrl+Z*

Scaling modifiers:

- Hold *Ctrl* while dragging a handle to preserve the image aspect ratio
- Hold *Shift* while dragging a handle to scale symmetrically around the image center
- Hold *Alt* while dragging a handle for finer, slower scaling

## Image rotation

While image transform mode is active, click the selected image to enter rotation mode. The corner handles change to curved rotation handles and a pivot marker appears.

In rotation mode:

- Drag a curved corner handle to rotate the image
- Drag the pivot marker to change the rotation center
- Hold *Ctrl* while rotating to snap the angle to the configured snap angle
- Hold *Shift* while rotating to keep the opposite corner fixed in place

For XVI images with an `xviRoot`, the pivot marker is shown but cannot be moved.

## Grid visibility

For XVI survey background images, a second checkbox controls whether the survey grid is displayed. Hiding the grid leaves the shots, stations, and sketch lines visible while removing the grid lines from the canvas. The grid visibility state is saved with the session.

## Image reordering

Click and drag the drag handle (⣿) of any image row to change its position in the list. The order of images in this list determines the rendering order on the canvas: images listed earlier are drawn below images listed later. Reordering is undoable with *Ctrl+Z*.

While dragging:

- The dragged row disappears from the list and a semi-transparent preview of it follows the cursor.
- A colored bar appears above the row where the dragged image will be inserted when released.

# Drag modifiers

Mapiah uses the same movement modifiers for background images, selected elements, and selected end/control points in single-line edit mode. These modifiers can be combined.

## Selected elements

When one or more elements are selected in selection mode:

- Press an *Arrow* key to move the selected elements by the configured nudge factor (`TH2Edit_NudgeFactor`), measured in canvas pixels
- Press *Shift+Arrow* to move by ten times the nudge factor
- Press *Alt+Arrow* to move by 1 screen pixel
- Press *Alt+Shift+Arrow* to move by 10 screen pixels
- Drag the selected elements to move them normally
- Hold *Alt* and drag anywhere on the canvas to move the current selection without changing the selection
- Hold *Ctrl* while dragging to constrain the move to the dominant horizontal or vertical direction
- Hold *Shift* while dragging to temporarily disable snapping

If the `TH2Edit_EnableElementTransforms` setting is enabled, the current selection also supports scaling, rotation, and mirroring:

- Drag any selection handle to scale the selected elements
- Hold *Ctrl* while dragging a selection handle to preserve the current aspect ratio
- Hold *Shift* while dragging a selection handle to scale symmetrically around the selection center
- Hold *Alt* while dragging a selection handle for finer, slower scaling
- Click the selected elements to switch from normal selection mode to element rotation mode
- In element rotation mode, drag a corner selection handle to rotate the selection
- Hold *Ctrl* while rotating to snap the angle to the configured snap angle
- Hold *Shift* while rotating to keep the opposite corner fixed in place
- Press *Alt+M* to mirror the selection horizontally
- Press *Ctrl+M* to mirror the selection vertically

## Single-line end/control points

When one or more end/control points are selected in line edit mode:

- Press an *Arrow* key to move the selected point or points by the configured nudge factor (`TH2Edit_NudgeFactor`), measured in canvas pixels
- Press *Shift+Arrow* to move by ten times the nudge factor
- Press *Alt+Arrow* to move by 1 screen pixel
- Press *Alt+Shift+Arrow* to move by 10 screen pixels
- Drag the selected point or points to move them normally
- Hold *Alt* and drag anywhere on the canvas to move the current point selection without changing the selection
- Hold *Ctrl* while dragging to constrain the move to the dominant horizontal or vertical direction
- Hold *Shift* while dragging to temporarily disable snapping

# Scraps

The project sidebar lists each `.th2` file's scraps in file order: the top row is drawn first. The former scrap dialog listed them in the opposite order. Open a file row with its chevron to see its scraps. Open canvas tabs that are not in the loaded project appear under **Open files outside the project**, even when no project is open.

Click a scrap row to make it active. You can also *Alt+click* an inactive scrap on the canvas, or press *Alt+K* to cycle through scraps. Right-click a scrap row for **Copy**, **Cut**, **Duplicate**, **Delete**, **Hide/Show**, **Options…**, and drawing-order actions. A selected scrap row's menu applies to all selected scraps; use *Ctrl+click* to select several. Copy places the scraps on the clipboard for *Ctrl+V*; Cut removes them as one undoable action; Duplicate copies them in the file as one undoable action. Delete is also undoable.

## Scrap visibility

Click the eye icon on a scrap row to show or hide it without opening its tab. A hidden scrap's label is dimmed, but its elements stay in the tree. The eye icon is absent when there is only one scrap. The last visible scrap cannot be hidden. The active scrap can't be hidden either: its eye icon is disabled until another scrap is made active. Activating a hidden scrap shows it. In text mode, clicking a scrap row only moves the text cursor; it doesn't activate the scrap.

Right-click a file row for **Hide all but active** or **Show all scraps**, and for **Add scrap**. The file visibility action does not open a tab.

## Scrap reordering

Drag a scrap row, use its drawing-order menu, or use the drawing-order shortcuts to move it in the stack. The changes are undoable with *Ctrl+Z*.

# Simplify lines

Bézier curves and straight line segments are simplified differently. To simplify lines, first select them. There can be other types of elements selected (points or areas) while simplifying lines. They will be untouched by the simplification process.

## Simplification methods

There are three line simplification methods available:

- **Keep original types (Ctrl+L)**: each line segment is simplified using its own type simplification algorithm.
- **Force Bézier (Ctrl+Alt+L)**: all line segments are, if necessary, first converted to Bézier curves and then simplified using the Bézier curve simplification algorithm.
- **Force straight (Ctrl+Shift+L)**: all line segments are, if necessary, first converted to straight lines and then simplified using the straight line simplification algorithm.

The **Interactive simplify lines dialog (Ctrl+Alt+Shift+L)** exposes the same parameters through a live preview. Use it to switch between the three methods and adjust the simplification intensity before committing the final result. Closing the dialog, saving it, or clicking outside the dialog keeps the current preview as a single undoable action. Cancelling restores the original selected lines without creating an undo entry.

### Bézier curve line segments

Each *Ctrl+[Alt]+L* press runs a round of line simplification. Mapiah uses an interactive algorithm to simplify Bézier curve line segments. It operates on canvas space. The initial tolerance (epsilon) is equivalent to 1.5 screen pixels. This value is converted to canvas coordinates. At each subsequent run the tolerance is increased by the same initial value.

### Mixed line segments

When a line contains both Bézier curve and straight line segments, Mapiah treats each part of the line separately. Each part containing only Bézier curve segments is simplified using the Bézier curve simplification algorithm. Each part containing only straight line segments is simplified using the straight line simplification algorithm.

### Straight line segments

Each *Ctrl+[Shift]+L* press runs a round of line simplification. Mapiah uses an interactive (non-recursive) version of the Ramer–Douglas–Peucker algorithm to simplify straight line segments. It operates on canvas space. The initial tolerance (epsilon) is equivalent to 1.5 screen pixels. This value is converted to canvas coordinates. At each subsequent run the tolerance is increased by the same initial value.

## Convert line segments

Mapiah can convert line segments between straight and Bézier types without changing the line selection itself.

**Shortcuts and buttons:**

- Press `J` or click the **Convert line segments to Bézier** button to convert to Bézier.
- Press `Shift+J` or click the **Convert line segments to straight** button to convert to straight.

**When is it available:**

- In line edit mode (*N*), the action is enabled when at least one selected point is not the start point of the line.
- In selection mode (*C*), the action is enabled when at least one line is selected.

**How it works:**

- In line edit mode, only the selected non-start line segments are converted.
- In selection mode, all selected lines are processed and every non-start segment is converted to the requested type.
- Segments that already match the requested type are left unchanged.
- Mixed lines can be converted in either direction.
- The operation can be undone with *Ctrl+Z*.

## Split line at selected points

While in line edit mode (*N*), select one or more internal line points and press *Ctrl+P* to split the line into multiple lines at those points.

**How it works:**

- The first new line contains all segments from the original line's start up to and including the first selected point.
- Each subsequent new line begins with a new straight segment whose endpoint is a copy of the last selected split point's position, followed by the original segments up to the next selected split point (or the end of the line).
- After the split, all newly created lines are selected.
- The operation can be undone with *Ctrl+Z*.

**IDs:** If the original line had an `-id` option (e.g. `wall1`), each new line receives a derived ID: `wall1-1`, `wall1-2`, and so on.

**Limitation:** Lines that are part of an area border cannot be split. A message is shown if you attempt to do so.

## Split lines at crossings

Press `Ctrl+Shift+X` to split the selected lines at every intersection with other selected lines in the same scrap.

**How it works:**

- For each selected line, every crossing point with any other selected lines is computed.
- Each crossing becomes a split point, producing multiple lines.
- New lines inherit options from the original; if the original line had an `-id` option (e.g. `wall1`), each new line receives a derived ID: `wall1-1`, `wall1-2`, and so on.
- After the split, all newly created lines are selected.
- The operation can be undone with `Ctrl+Z`.
- **Limitation:** Lines that are part of an area border cannot be split. A message is shown if you attempt to do so.

## Join lines at coinciding extremities

Press `Ctrl+J` (or click the **Join lines at coinciding extremities** button) to join selected lines whose start/end extremities coincide.

**When is it available:** The action is enabled when at least two lines are selected.

**How it works:**

- The operation checks selected lines for extremities that coincide using a tolerance equivalent to 3 screen pixels (converted to canvas coordinates at execution time according to current zoom).
- A single run can produce multiple joined lines when selected lines form multiple independent groups.
- For each resulting joined line, the line type and options from the first selected line in that group are preserved.
- Line segments are reoriented as needed so the joined geometry is continuous; Bézier segments are reversed preserving their visual shape.
- The resulting joined lines are selected.
- The operation can be undone with *Ctrl+Z*.

If no coinciding extremities are found, Mapiah shows a message and performs no changes.

## Merge areas

Press *Ctrl+Alt+M* (or click the **Merge areas** button) to merge the border lines of the selected areas into the fewest possible closed lines, replacing the selected areas with a single merged area.

**When is it available:** The action is enabled when the total number of distinct border lines across the selected areas is two or more. Areas can be selected directly, or indirectly by selecting one or more of their border lines. This covers two common scenarios:

- One area that already has more than one border line (`THAreaBorderTHID`).
- Two or more areas selected, whether by selecting the areas themselves or their border lines.

**How it works:**

1. Mapiah collects all selected areas plus any areas referenced by selected border lines, then gathers their distinct border lines (LTSAs — Lines To be merged from Selected Areas).
2. Any LTSA that is not already closed (last point ≠ first point) is automatically closed with a straight segment.
3. The LTSAs are grouped by mergeability. Lines belong to the same group when they cross each other or share a geometrically identical segment; isolated lines form singleton groups.
4. For each group containing more than one line, all segments are assembled into a single merged line using a bounding-path algorithm that traces the outer boundary.
5. A single new area is created, referencing all merged lines as its borders.
6. The resulting merged area is selected after the operation.
7. The operation can be undone with *Ctrl+Z*.

**Options and IDs:**

- The new area inherits all options (type, subtype, etc.) from the first selected area.
- Each merged line inherits all options from the first LTSA in its group.
- Explicit Therion IDs (`-id`) are preserved where they existed on the canonical area and canonical line of a group. Lines and areas without explicit IDs receive auto-generated IDs so they can be properly referenced as area borders.

If segments outside the outer boundary are detected during the merge, Mapiah shows an error message and performs no changes.

## Hide elements

Press *Ctrl+H* to temporarily hide elements on the canvas without removing them from the file.

**When elements are selected:**

- The selected elements are added to the hidden list and deselected.
- Hidden elements are no longer drawn on the canvas and cannot be clicked or selected.

**When no elements are selected:**

- All hidden elements are made visible again.

Hidden elements are a temporary canvas-only state: they are not saved to the file and are always restored when the file is reopened.

## Search and select

The search and select dialog allows you to find and select elements in the current scrap based on their characteristics. Open it by clicking the 

button in the top right corner.

The dialog has three collapsible sections: **Points**, **Lines**, and **Areas**. Enable a section by checking its checkbox. Each enabled section offers filtering criteria:

- **All**: selects all elements of that type in the current scrap. Enabling this disables the other criteria.
- **By ID**: filters elements whose Therion ID contains the entered text (case-insensitive partial match).
- **By subtype**: filters elements by their subtype. Select known subtypes from chips and/or enter free text for unknown subtypes. Available for points and lines only.
- **By type**: filters elements by their type. Select one or more types from the available chips. Unknown types found in the current scrap are also listed.
- **By option**: filters elements by whether specific options are set or not. Each option can be set to *Undefined* (ignored), *Set* (element must have the option), or *Unset* (element must not have the option).
- **By line segment option** *(lines only)*: filters lines by whether any of their line segments have specific options set or not. Each option can be set to *Undefined* (ignored), *Set* (at least one segment of the line must have the option), or *Unset* (no segment of the line may have the option).

When multiple criteria are enabled within a section, an element must match **all** of them (AND logic). When multiple sections are enabled, matching elements from any section are included (OR logic).

A status bar at the bottom shows the number of elements matching the current criteria, updated live as you change the filters.

**Action buttons:**

- **Set selection**: replaces the current selection with the matching elements.
- **Add to selection**: adds the matching elements to the current selection.
- **Remove from selection**: removes the matching elements from the current selection.
- **Cancel**: closes the dialog without changing the selection.
- **Reset**: resets all criteria without closing the dialog.

## Snap

There are several snap options available that can be controled on the window presented when the button  is pressed:

- XVI file snap (zero or more options can be selected):
  - **Grid lines**: snap to grid lines.
  - **Grid line intersections**: snap to grid line intersections on the XVI file.
  - **Shots**: snap to shot start and end points on the XVI file.
  - **Sketch lines**: snap to sketch line line points on the XVI file.
  - **Stations**: snap to stations defined on the XVI file.
- Point snap (single option):
  - **None**: no snaping to points in the TH2 file.
  - **Points**: snap to all defined points in the TH2 file.
  - **Points by type**: snap only to the selected point types in the TH2 file.
  - **Snap to inactive scraps**: also use matching points from visible inactive scraps.
- Line point snap (single option):

- **Lines point**: snap to all line points in the TH2 file.
- **Lines point by type**: snap only to the line points of the line types selcted.
- **None**: no snaping to line points.
- **Snap to inactive scraps**: also use matching line points from visible inactive scraps.

## Zoom and panning

The TH2 file view can be zoomed in and out using the zoom buttons or the mouse wheel. The view can also be panned by right clicking and dragging the mouse. *Ctrl+mouse wheel* pans vertically, *Shift+mouse wheel* pans horizontally.

# Workspace keyboard shortcuts

*Note: Mapiah treats the Ctrl and Meta (Command on macOS) keys as interchangeable. Shortcuts below show "Ctrl" for brevity.*

| Function | Shortcut |
|---|---|
| Create file | Ctrl+N or Ctrl+Shift+N |
| Help | F1 |
| Keyboard shortcuts | Ctrl+K |
| Open project | Ctrl+O or Ctrl+Shift+O |
| Open project and run Therion when no project is loaded | Ctrl+T |
| Run/rerun Therion for the current project | T |

## Text editor

These shortcuts apply when a `thconfig` or `.th` text tab is active.

| Function | Shortcut |
|---|---|
| Find and replace in the active text file | Ctrl+F |
| Save the active text file | Ctrl+S |

`Ctrl/Cmd+T` is unavailable for switching projects once a project is loaded.

# Drawing keyboard shortcuts

*Note: Mapiah treats the Ctrl and Meta (Command on macOS) keys as interchangeable. Shortcuts below show "Ctrl" for brevity.*

| Function | Shortcut |
|---|---|
| Add image | I |
| Add line segment between selected line segments | / |
| Bring selected elements forward | Ctrl+] |
| Bring selected elements to front | Ctrl+Shift+] |
| Change active scrap | Alt+K |
| Change active scrap by clicking inactive scrap | Alt+click inactive scrap |
| Change images | Alt+I |
| Change unfinished drawing segment from curve to line | Shift+L |
| Change unfinished drawing segment from line to curve | Shift+U |
| Open project and run Therion when no project is loaded | Ctrl+T |
| Click area-border line: cycle border-line selection | Ctrl+click |
| Click area-border line: select area only | Ctrl+Alt+click |
| Convert selected line segments to Bézier curve | J |
| Convert selected line segments to straight | Shift+J |
| Constrain a new line-drawing node to the snap angle while drawing | Shift+click |
| Copy selected elements | Ctrl+C |
| Create area | A |
| Create freehand line | F |
| Create line | L |
| Create map connection line from point X-Section to Station | Ctrl+Alt+X |
| Create point | P |
| Create scrap | K |
| Cut selected elements | Ctrl+X |
| Delete an unfinished path while drawing | Esc |
| Delete last created node while drawing | Del or Backspace |
| Default options (no elements selected) | O |
| Delete point, line, area or line segment | Del or Backspace |
| Deselect all | Escape |
| Duplicate selected elements | Ctrl+D |
| Find and replace text across multiple project files | Ctrl+Shift+F |

| Function | Shortcut |
| --- | --- |
| Find text in the active text file | Ctrl+F |
| Force option setting on non-supporting types | Ctrl+click on option setting |
| Help | F1 |
| Hide selected elements / Show all hidden elements | Ctrl+H |
| Join lines at coincident start/end line points | Ctrl+J |
| Keyboard shortcuts | Ctrl+K |
| Line or line segment: enter line edit | Double-click |
| Node/line edit | N |
| Merge areas | Ctrl+Alt+M |
| Mirror selected image / elements horizontally | Alt+M |
| Mirror selected image / elements vertically | Ctrl+M |
| Move last created line-drawing node by 1 screen pixel | Alt+Arrow |
| Move last created line-drawing node by 10 screen pixels | Alt+Shift+Arrow |
| Move last created line-drawing node by nudge factor | Arrow |
| Move last created line-drawing node by 10 times the nudge factor | Shift+Arrow |
| Move selected image / elements / end-control points by 1 screen pixel | Alt+Arrow |
| Move selected image / elements / end-control points by 10 screen pixels | Alt+Shift+Arrow |
| Move selected image / elements / end-control points from anywhere | Alt+drag |
| Move selected image / elements / end-control points constrained | Ctrl+drag |
| Move selected image / elements / end-control points without snap | Shift+drag |
| Move selected image / elements / end-control points by nudge factor | Arrow |
| Move selected image / elements / end-control points by 10 times the nudge factor | Shift+Arrow |
| Options edit (elements selected) | O |
| Paste elements | Ctrl+V |
| Project tree: select an element range | Shift+click |
| Project tree: toggle an element or scrap | Ctrl+click |
| Rotate image | drag image rotate handle |
| Rotate image with angle snapping | Ctrl+drag image rotate handle |
| Rotate image while keeping opposite corner fixed | Shift+drag image rotate handle |
| Rotate selected elements | drag selected-elements corner handle |
| Rotate selected elements with angle snapping | Ctrl+drag selected-elements corner handle |
| Rotate selected elements while keeping opposite corner fixed | Shift+drag selected-elements corner handle |
| Scale image from handle symmetrically | Shift+drag image handle |
| Scale image from handle with aspect ratio preserved | Ctrl+drag image handle |
| Scale image from handle with finer control | Alt+drag image handle |

| Function | Shortcut |
| --- | --- |
| Scale selected elements from handle symmetrically | Shift+drag selected-elements handle |
| Scale selected elements from handle with aspect ratio preserved | Ctrl+drag selected-elements handle |
| Scale selected elements from handle with finer control | Alt+drag selected-elements handle |
| Redo | Ctrl+Shift+Z |
| Set a new control point constrained to the snap angle while drawing | Shift+drag |
| Run Therion | T |
| Save file | Ctrl+S |
| Save file as | Ctrl+Shift+S |
| Set next xTherion cubic smooth control point separately while drawing | Alt+drag after starting drag |
| Select all | Ctrl+A |
| Select none | Esc |
| Select tool | C |
| Send selected elements backward | Ctrl+[ |
| Send selected elements to back | Ctrl+Shift+[ |
| Simplify line - interactive dialog | Ctrl+Alt+Shift+L |
| Simplify line | Ctrl+L |
| Simplify line - to Bézier curve line segments | Ctrl+Alt+L |
| Simplify line - to straight line segments | Ctrl+Shift+L |
| Smooth line point | S |
| Split line at selected line points | Ctrl+P |
| Split selected lines at crossings | Ctrl+Shift+X |
| Switch between canvas and text mode (`.th2`) | F2 |
| Toggle all grids visibility | Ctrl+G |
| Toggle all images visibility | Ctrl+I |
| Toggle border line option | B |
| Toggle direction ticks on non-selected lines | Ctrl+Alt+R |
| Toggle reverse line option | R |
| Toggle visibility option | V |
| Undo | Ctrl+Z |
| Zoom 1:1 | 1 |
| Zoom file | 4 |
| Zoom in | + |
| Zoom out | - |
| Zoom scrap | 3 |
| Zoom selection | 2 |

| Function | Shortcut |
| --- | --- |
| Zoom selection window | 5 |

*In a `.th2` tab's text mode, Ctrl+Z and Ctrl+Shift+Z undo and redo canvas edits until the first text edit; from then on, they act on the text only (see the TH2 file edit help, "Text mode").*

## Text file search

- **Ctrl+F** searches the text file open in the active editor tab, including a `.th2` tab in text mode.
- **Ctrl+Shift+F** opens multi-file search in the project sidebar.
  - **Open text tabs** scope searches every open `thconfig/.th` text tab; **Project files** scope searches every writable `thconfig/.th` file in the loaded project, including files that are not open.
  - `.th2` drawings are never searched, not even in text mode.
  - A text tab that is open but is not part of the loaded project is searchable and you can navigate to its matches, but it is marked *Search only* and Replace All never modifies or saves it.
  - Activating a result opens or focuses the file tab, reveals the file in the project tree, and selects the exact match. If the file changed since the search ran, the search is refreshed and the nearest current match is selected.
  - Replace All asks for confirmation, reports how many matches and files are affected (and how many are excluded), then replaces and saves the affected project files. There is no single-step undo across files.

# Settings

This page allows you to configure Mapiah's settings. Settings are grouped into sections displayed as cards.

*Note: Mapiah treats the Ctrl and Meta (Command on macOS) keys as interchangeable. Shortcut mentions below use "Ctrl" for brevity.*

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## Using settings

Changes made on the settings page are not applied immediately. They are held as draft values until you explicitly save or apply them.

### Resetting a single setting

Each setting has a reset button (↺) on its right side. Clicking it reverts that setting to its default value in the draft, without saving.

### Buttons

- **Save & Close**: applies all draft changes and closes the settings page. If any field contains an invalid value, the save is blocked and the invalid fields are highlighted.
- **Apply**: applies all draft changes but keeps the settings page open.
- **Cancel**: discards all draft changes and closes the settings page.
- **Reset all settings**: reverts all settings to their default values in the draft, without saving.

## Main section

### Language

Selects the language used throughout the app interface. The default option follows the system locale. Changing this setting takes effect after applying and reopening any currently open pages.

## Text editor section

### Font size

Controls the font size (in pixels) of the text editors used for `.th`, `thconfig` and `.th2` files, and of their line numbers. The default is `13`. Values are limited to the range from `6` to `48`. The change takes effect in open editors as soon as it is applied.

## TH2 edit section

### Enable element transforms

When enabled, selected elements gain scale, rotate, and mirror actions in selection mode. Selection handles can be dragged to scale, clicking the selected elements enters rotation mode, and *H* / *V* mirror the current selection. The default is disabled (`false`).

### Enable special border for ID set

When enabled (default), elements with THID/ID set are painted with a dedicated special border on the canvas. Disable this if you prefer those elements to use only their normal symbol styling.

### Enable special border for slope line without l-size

When enabled (default), slope lines whose line segments do not define any `l-size` are painted with a dedicated special border on the canvas. This can help you spot slope lines that still need `l-size` information.

### Enable special border for visibility off

When enabled (default), elements with `visibility off` are painted with a dedicated special border on the canvas. Disable this if you prefer invisible elements not to receive extra visual emphasis while editing.

## Line thickness

Controls the visual thickness (in pixels) of lines drawn on the canvas. This is a display-only setting and does not affect the data stored in the TH2 file.

## New line creation method

Controls the behavior when creating a new line segment by clicking and dragging:

- **Mapiah quadratic**: the drag position is used as the single control point of a quadratic Bézier curve approximation. Hold *Shift* while dragging to constrain that control point to the nearest multiple of the configured snap angle relative to the shared node.
- **xTherion cubic smooth**: the drag position becomes the next segment's future control point; the current segment's other control point is mirrored around the shared endpoint. Hold *Ctrl* while dragging to lock the mirrored control point at a fixed distance, hold *Shift* while dragging to constrain that future control point to the nearest multiple of the configured snap angle relative to the shared node, hold *Shift* while setting a node to constrain it to the nearest multiple of the configured snap angle relative to the previous node, and hold *Alt* after starting the drag to redefine only the next segment's future control point. In this mode, *Shift* can also be combined with *Alt*. It tries to reproduce XTherion behaviour.

While drawing with either method, the last created node can also be nudged with *Arrow*, *Shift+Arrow*, *Alt+Arrow*, and *Alt+Shift+Arrow*.

## Point radius

Controls the visual radius (in pixels) of points drawn on the canvas. This is a display-only setting and does not affect the data stored in the TH2 file.

## Selection tolerance

Controls how close the mouse cursor must be to an element (in pixels) for it to be considered clicked and selected. Increasing this value makes elements easier to click.

## Show direction ticks on non-selected lines

When enabled, direction ticks are drawn on all lines in the active scrap, regardless of selection. When disabled (default), only selected lines show direction ticks. Can also be toggled with **Ctrl+Alt+R**.

## Show line points

When enabled (default), every line point is drawn as a black circle whose diameter is 1.5 times the configured line thickness. The points remain visible on selected and non-selected lines, including lines in inactive scraps. Disable this setting to draw lines without these persistent points.

## Snap angle

Controls the angular increment (in degrees) used when snapping image rotation and new-line drawing constraints. While rotating an image, hold *Ctrl* to snap the angle to multiples of this value. While drawing a line, hold *Shift* to constrain a new node or a dragged control point to multiples of this value. Set it to `0` to disable snapping even while the modifier is held.

### Symbol unit

Controls the base size (in pixels) of Therion symbol-set renderings, i.e. the u unit used throughout Therion's symbol definitions. Only affects the canvas when a Therion visualization method (not **Mapiah placeholder**) is selected. This is a display-only setting and does not affect the data stored in the TH2 file.

### Visualization method

Selects how TH2 point, line, and area symbols are drawn. **Therion (default)** renders exactly what a real Therion run without a `symbol-set` layout option produces (Therion's own `thTrans.mp` default translation table) and is the default for new installs and setting resets. **Therion UIS** is Mapiah's equivalent of Therion's `symbol-set UIS` layout option: it overrides the default to UIS wherever a UIS symbol exists, leaving `thTrans.mp`'s default in force everywhere else. **Mapiah placeholder** preserves the existing geometric rendering instead of any Therion symbol set.

## Therion section

### Debug log 1

When enabled, Mapiah shows extra Therion startup and run diagnostics. The default is disabled (`false`).

### Therion executable path

The full path to the Therion executable on your system. Required for the *Run Therion* feature to work. Click the folder icon or tap the field to open a file picker and navigate to the Therion binary.

### Therion run parameters

Optional extra command-line options passed to Therion on every run (e.g. `-d` for debug mode, `-q` for quiet mode). Multiple options can be entered space-separated. The value is also editable directly in the Run Therion dialog and can be preset via the `--therion_run_parameters` Mapiah command-line argument. Default is empty (no extra options).

# Run Therion

This dialog runs Therion with the loaded project's root configuration file and shows its output in real time.

## Status

Shows the current state of the Therion run:

- **Running** — Therion is currently executing.
- **Ok** — Therion finished with no warnings or errors.
- **Warning** — Therion finished but reported one or more warnings.
- **Error** — Therion finished with one or more errors, or could not be started.

## Therion run parameters

Optional extra command-line options passed to Therion on every run (e.g. `-d` for debug mode). The value is saved as a persistent setting and can also be set via:

- The **Settings page** (`Therion_RunParameters` field).
- The Mapiah `--therion_run_parameters` command-line argument (see the Main page help for details).

## Broken files warning

When you run the loaded project and some of its `.th2` files that Mapiah has already loaded are broken on disk, which is what Therion reads, a warning above the output lists their paths. Therion may fail because of them, but the run starts anyway. The paths are selectable text, and a long list scrolls.

- Only files of the loaded project that is being run are listed, and only those already loaded, by expanding them in the project tree or opening them in a tab. Mapiah does not read other files to look for problems.
- No warning appears when you run a configuration file that is not the loaded project, or right after choosing a project to run, before it has loaded.
- A file fixed in text mode but not saved yet is still listed, marked "(fix not saved)".
- The list is taken when the dialog opens. After a fix is saved, or after fixing a file elsewhere and reloading it, the next run no longer lists it.

## Output

The full text output produced by Therion during the run. After the run finishes, the Therion log file is appended, followed by the start and end times.

- **Warning** and **Error** keywords are highlighted in color.
- The output area is scrollable and its text is selectable.
- Clicking an item in the issues list (see below) scrolls the output to the corresponding line.
- A diagnostic with a recognized project file and source line can open that text tab at the line. Diagnostics without a file or line remain visible here but cannot be mapped to a tree line.

## Elapsed time

Shows the time elapsed since the run started. Updates live every second while Therion is running, then freezes when the run finishes.

## Issues list

When Therion reports warnings or errors, they appear as a scrollable list below the output area. Clicking any item scrolls the output to that line.

## Buttons and keyboard shortcuts

- **Rerun Therion** (keyboard: **T**) — runs Therion again with the same THConfig and current run parameters. Only enabled when Therion is not running.
- **Close** (keyboard: **Escape**) — stops any in-progress Therion run and closes the dialog.
- **Ctrl/Cmd+T** — when no project is loaded, opens a project and starts Therion. It is unavailable for switching projects once a project is loaded or while this run dialog is active.

Compiler diagnostics remain until a subsequent run replaces them; editing a file does not by itself prove that a compiler error is fixed.

# Telemetry

# Telemetry — Anonymous Usage Data

Mapiah can optionally collect anonymous, aggregated usage data to help prioritize development and understand how the application is used.

## What is collected

Each day that you use Mapiah, a single aggregated record may be sent containing:

| Field | Description |
|---|---|
| Date | UTC date (e.g. 2026-03-20) |
| OS type | linux, macos, or windows |
| OS version | raw version string from the operating system |
| Linux distro | distribution name (Linux only, e.g. "Fedora Linux 43") |
| Window manager | desktop environment (Linux only, e.g. "KDE") |
| Mapiah version | the version of Mapiah you are running |
| Build type | AppImage, Flatpak, or Other |
| TH2 files (unique) | number of distinct TH2 files opened that day |
| TH2 open count | total number of times a TH2 file was opened |
| TH2 time (minutes) | total minutes with at least one TH2 file open |
| THConfig files (unique) | number of distinct THConfig files used |
| Therion run count | number of times Therion was run |
| Therion time (seconds) | total seconds spent running Therion |

## What is NOT collected

- Your name, email, or any account information
- Device identifiers, IP addresses, or location
- File names or file paths
- The content of any cave survey or TH2 file
- Any information that could identify you or your computer

## How data is handled

Data is collected locally throughout the day and only the previous day's aggregated record is ever transmitted. No partial or real-time data leaves your computer. Records are sent via HTTPS to `api.mapiah.org`. If a send fails (no network, server down), Mapiah retries automatically every 15 minutes.

When you first accept or decline telemetry, an immediate notification (containing no identifying information) is sent to the server so the anonymous opt-in or opt-out counter can be incremented. This lets Mapiah understand overall consent trends without any user-level tracking.

## How to change your choice

You can enable or disable telemetry at any time on the *Settings* page under the **Main** section. The toggle labeled "Share anonymous usage data" controls this setting. You can also click "Review telemetry details and consent" on the Settings page to see this dialog again.

When you opt out, all locally stored telemetry data is deleted immediately and an anonymous opt-out notification is sent to the server. When you opt back in, an anonymous opt-in notification is sent.

## Open source transparency

As with all open source software, you can review the source code to see exactly how telemetry works.

The relevant code is in `src/telemetry/` and `src/controllers/MPTelemetryController.cpp`.

Telemetry data is stored locally in Mapiah's internal settings file, and sent to the server via HTTPS POST requests to `https://api.mapiah.org/telemetry`.

# Telemetry consent

Mapiah is an open source software developed by a very small team on a voluntary basis.

You can help improve Mapiah allowing it to collect **anonymous, aggregated usage data**.

**No personal information, file contents, or identifying data is ever collected or transmitted.**

You can change this setting at any time on the *Settings* page.

# Therion installation

# No Therion found

Mapiah needs access to a previously installed Therion to work. If you see this message, it means that Mapiah cannot find a Therion installation on your system.

If you don´t have Therion installed, you can download it from the Therion download page.

If you already have Therion installed, make sure that it is working properly.

If still Mapiah can't find Therion, you can specify the path to the Therion executable in Mapiah settings.

# Flatpak information

# Update available

Newer versions of Mapiah are available, but this Flathub build is not upgraded anymore.

Update instrcutions (on the terminal):

```
# Remove this version of Mapiah.
flatpak remove io.github.rsevero.mapiah
# Install the latest Flatpak version of Mapiah from the official repository.
flatpak install --user --from
https://flatpak.mapiah.org/org.mapiah.mapiah.flatpakref
```

If you want, you can also check more info on the Mapiah page on Flathub: Mapiah on Flathub

## Rationale

The Mapiah Flathub build is no longer maintained due to the following factors:

- The build process is excessively slow, resource-intensive, and involves multiple manual steps;
- Flathub's restrictions on external applications require Mapiah to bundle Therion and its dependencies, increasing the package size from roughly 14MB to over 950MB;
- Bundled Therion is significantly slower than the native one.